ReleaseProof

FIRMWARE MIGRATION ASSURANCE · INTELLIGENT INDUSTRY

A firmware can boot and still be impossible to upgrade safely.

Help industrial IoT manufacturers verify that an update can **reach installed devices**—not just boot on a new one—before rollout. AMD-hosted AI plans and explains. **The engine decides.**

2 live configurations · Nordic nRF52840 | wolfBoot & MCUboot | ECDSA P-256 → ML-DSA | preflight, not OTA delivery | every artifact sha256-hashed



Values copied from a real run of the app (Renode emulation, 17 Oct 2026).

THE PROBLEM

A factory's sensors must remain updatable—not just functional on day one

OEM
OUR FIRST BUYER

Wireless condition-monitoring manufacturers

Firmware leads and product-security managers need to keep installed sensors maintainable, protecting the monitoring that supports industrial maintenance.

nRF52840
REAL HARDWARE FIT

NI MON-10411 · Lynxemi · Atomation

Nordic documents this SoC in industrial wireless sensors. Our demo uses the same MCU family, **not** these vendors' proprietary firmware or bootloaders.[1]

Field
THE BUSINESS RISK

A rejected update can require physical intervention

Hard-to-reach sensors turn a bootloader mismatch into engineering rework, field visits and potential gaps in monitoring. No customer incident or savings are claimed.

CRA
CONTEXT, NOT PROOF

Secure-update practices and evidence matter

Main CRA obligations: 11 Dec 2027; reporting: 11 Sep 2026. Scope, exclusions and transition rules apply. **No general ML-DSA mandate or CRA certification claim.**[2]

The trap: "boots" ≠ "migrates"

LAB TEST: CANDIDATE STATE

ML-DSA-87 bootloader + ML-DSA-87 signed image → Boots

```
UART: version frame 2a 00 00 00 01 → v1 running
```

FIELD REALITY: THE TRANSITION

Deployed ECDSA P-256 bootloader
no self-update on this port

← Proposed ML-DSA-87 update → Rejected

```
UART: "Update type invalid 0xB01≠0x200"; v1 keeps running
```

A passing boot test does not prove a deployable update. On this profile, the installed bootloader cannot get there through the proposed path.

Both lanes were measured live by ReleaseProof (wolfBoot @4c79eee, Renode 1.16.1, run RP-20261017T124806Z-a39610).

[1] Nordic Semiconductor news: NI (Sep 2019), Lynxemi (Apr 2020), Atomation (Jun 2023). Hardware relevance only; no customer relationship or commercial image support. [2] European Commission CRA summary/reporting. Source links in pitch/submission.md.

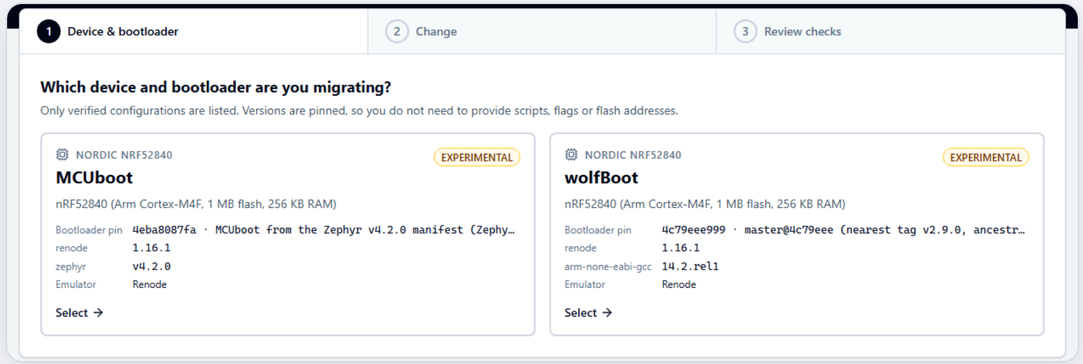
THE SOLUTION, LIVE

Choose a supported configuration and inspect the migration path before rollout

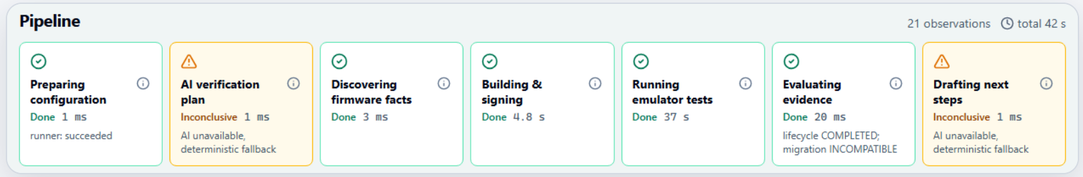
1 Pick a supported configuration
Device + bootloader with pinned commits and toolchains. No scripts, flags or flash addresses to type.

2 Pick the change
For example ECDSA P-256 → ML-DSA-87. Review the required checks and the ones the AI may add.

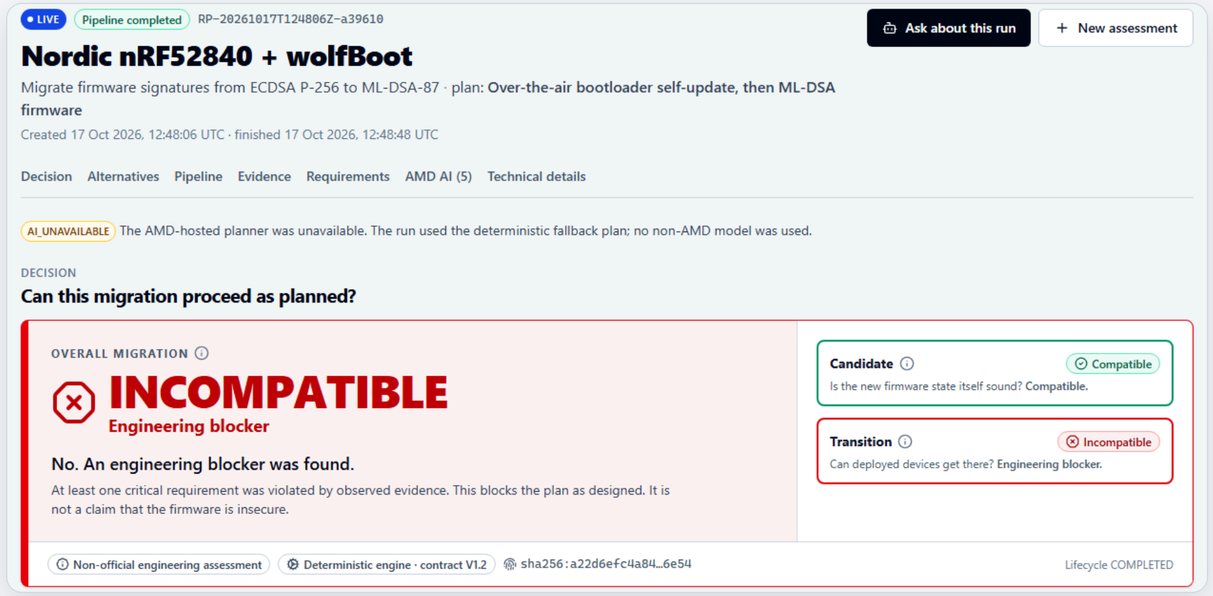
3 Run: it really executes
Builds and signs with ephemeral keys, then boots and updates in Renode. Positive and negative tests feed the V1.2 verdict, canonical hash and manifest.



Historical screenshot: wizard with pinned versions. Current-source acceptance must be rerun before the final submission.



Recorded pipeline: 7 stages, 21 observations. This 42 s historical build/emulation run is not an AI-enabled end-to-end latency guarantee.



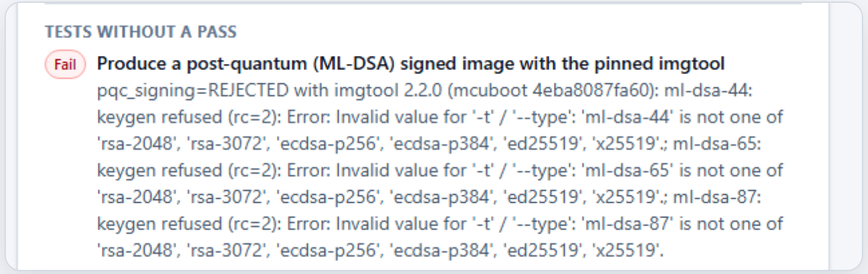
Historical LIVE decision: candidate and transition judged separately by the engine. Run `RP-20261017T124806Z-a39610`. Evidence links to hashed UART captures, ELF files, signed images and Renode scripts. This screenshot is not proof of AMD-enabled inference.

TWO CONFIGURATIONS, BOTH RUNNING END TO END

Same chip, two bootloaders, and two different reasons the migration fails

	nRF52840 + wolfBoot change: ECDSA P-256 → ML-DSA-87	**nRF52840 + MCUboot (Zephyr)** change: ECDSA P-256 → ML-DSA
Pinned stack	`wolfBoot 4c79eee · wolfSSL 4aa1ad7 · Arm GCC 14.2.rel1`	`MCUboot 4eba8087 · Zephyr v4.2.0 · SDK 0.17.4 · imgtool 2.2.0`
Emulation	Renode 1.16.1, nRF52840; SVD baked in and sha256-pinned	Renode 1.16.1 with ReleaseProof's own NVMC flash model (MIT), swap-using-move
Candidate check	ML-DSA-87 bootloader and image built and **booted** (E02); signed image 12 876 B fits the 262 144 B slot	PQC signing attempt with the pinned tool (E02): **imgtool refuses ml-dsa-44/65/87** (measured)
Field-update path	Source inspection (E11): no nRF52 self-update path (`RAM_CODE=0`)	Partition map + Kconfig (E11): single-stage MCUboot, no bootloader slot
Negative tests	Tampered image rejected (E03) · deployed ECDSA bootloader rejects the ML-DSA update (X01) · candidate rejects an ECDSA image (P01)	Tampered image rejected (E03) · update signed with another key rejected (X01)
Engine verdict	Candidate COMPATIBLE · Transition INCOMPATIBLE REQ-BOOT-S0S1 violated · run a39610 · 8/8 tests · 61 artifacts	Candidate INCOMPATIBLE · Transition INCOMPATIBLE REQ-PQC-VERIFIER + REQ-BL-FIELD-UPDATE · run 7682fc · 7/7 tests · 50 artifacts

Why the checks differ: the test semantics are bootloader-agnostic (E01 current boots, E02 candidate, E03 auth-negative, X01 cross-transition, E05 layout, E11 field-update capability). Each profile maps them to its own build and emulation templates. **Runners measure; only the engine decides.** A missing observation becomes `UNKNOWN`, never a pass.



Historical runs a39610 / 7682fc and screenshot, 17 Oct 2026. Profiles and runner READMEs document the scope. A fresh strict acceptance must identify the same backend/runner release before submission.

THE AGENTIC LOOP

The AI proposes, the emulator measures, the engine decides, and a human chooses the next step

AMD-HOSTED AI

① Plan tests

Picks and orders the optional registered checks. Required checks always run, and unknown templates are blocked and shown.

→

RUNNER

② Execute for real

Pinned build, ephemeral keys, Renode boot and update, UART capture. No network access, and every artifact is hashed.

→

V1.2 ENGINE

③ Deterministic verdict

Candidate, transition and migration verdicts plus an explanation graph and a canonical hash. Golden-hash regression tests guard it.

→

AMD-HOSTED AI

④ Explain + alternatives

Explanations must cite real evidence IDs. Alternatives come only from the profile's allowed set and are labelled PROPOSAL.

→

HUMAN + ENGINE

⑤ Re-evaluate

A human picks an alternative. That creates a **new** run with a parent reference, and history is never changed.

↺ the re-evaluation feeds back into ③ or, when new tests are needed, starts a new LIVE run at ②

`LIVE · RP-20261017T124806Z-a39610 · bootloader_self_update`

INCOMPATIBLE

REQ-BOOT-S0S1 is violated: the deployed wolfBoot cannot replace itself over the air on nRF52840.

→ human clicks "Evaluate this plan"

`REEVALUATION · RP-20261017T124931Z-72d0b1 · field_service_reflash`

INDETERMINATE

The blocker is gone, but recovery from an interrupted update (INV-I4) was never observed, so the result is an honest unknown and not a pass. The parent's hash is unchanged.

Schema-validated structured output; bad JSON is repaired once, otherwise rejected

Citation check: structured claims cite allowed IDs; buffered chat checks ID existence

Verdict-phase check: explicit labels cannot swap candidate success into migration success

Allow-listed AMD host and no fallback provider. If the AI is down: `AI_UNAVAILABLE` and the deterministic plan runs

Tool output is untrusted: UART and log text is wrapped against prompt injection

No shell from the AI: it can only choose allow-listed templates

The example run pair is historical. Show actual AMD traces for the demo run. Guardrails check IDs/explicit labels, not semantic truth of every sentence; human engineering review remains required.

ARCHITECTURE ON AMD

AMD powers supervised planning and explanations—not pass/fail decisions

DOCKER COMPOSE · CLOUD TARGET: AWS C7A / AMD EPYC

CLOUD DEPLOYMENT NOT ATTESTED

Browser · Next.js UI
wizard · live run (SSE) · evidence · alternatives · AMD panel

Caddy
auto-TLS edge · /api → api · / → web

FastAPI api
orchestrator state machine · queue · SQLite · artifacts · manifests

↓ HTTP / NDJSON (internal network, no egress)

runner-wolfboot
wolfBoot + wolfSSL · Arm GCC 14.2 · keytools · Renode 1.16.1 · own SVD

runner-mcuboot
Zephyr v4.2.0 sysbuild · MCUboot · imgtool 2.2.0 · Renode + NVMC model

↓ typed observations + artifact sha256

Assessment adapters
observations → V1.2 assessment; missing fact ⇒ UNKNOWN / gap

V1.2 semantic engine
validate · evaluate · explanation graph · canonical hash

AI gateway
planner · explainer · alternatives · assistant; schema, citation and verdict-echo guards; traced

↓ persisted per run

Run record (immutable)
verdicts · explanation graph · gap plan · dossier · `manifest.json` with inputs, pins, artifact sha256 and result hash · AI traces

Run lifecycle: PREPARING → AI_PLAN → DISCOVERY → BUILD_SIGN → EMULATION_TESTS → EVALUATION → NEXT_STEPS. If a runner is killed or crashes, the run is FAILED and no verdict is issued. Secrets live only in the api container.

AMD EVIDENCE

AMD Token Factory · managed GPU undisclosed
OpenAI-compatible, operator allow-listed host, reached through API egress only

HOST CPU	Deployment target: AMD EPYC 4th gen (EC2 c7a.2xlarge, 8 vCPU); cloud deployment not yet evidenced
GPU	Managed GPU undisclosed; labels are not attestation
MODEL	Google Gemma-4-31B-IT on AMD
SERVING	AMD Token Factory managed API
SAMPLE	2-iteration run: 12/14 calls OK, 2 timeouts, schema-valid 1.0. Not a performance benchmark.
TRUST BOUNDARY	Buffered chat and reports check citation IDs and explicit verdict phases. This is not semantic proof of every sentence.
UNKNOWNS	Emulation ≠ silicon. Interrupted-update recovery stays UNKNOWN; no power-loss safety claim.

Retained Token Factory smoke is historical. Fresh traces must be correlated to the run shown in a live demo. Runner crashes produce FAILED, not a firmware verdict. Neither provider labels nor artifact hashes attest managed GPU identity.

MARKET AND BUSINESS MODEL

Start with industrial sensor OEMs: verify one migration before risking a rollout

ILLUSTRATIVE ECONOMICS—NOT OBSERVED ROI

5,000
hypothetical sensors

A fleet used only to illustrate exposure. ReleaseProof evaluates a configuration and transition, not each node individually.

5% × €250
hypothetical intervention

Assume 250 devices require physical service, at €250 per visit. These assumptions are not a measured failure rate or market price.

€62,500
scenario cost—not savings

5,000 × 5% × €250. We have not demonstrated that ReleaseProof avoids this cost; validate with design partners.

What a pilot should measure

Pre-rollout blockers found, engineering hours saved, validated field visits avoided, and cost per supported profile/run.

BUYER: EMBEDDED ENGINEERING / FIRMWARE / PRODUCT SECURITY

Configuration-profile subscription — RECURRING

Per device + bootloader + toolchain profile, kept pinned and re-validated as upstream changes. Covers wolfBoot, MCUboot and future bootloader families.

Assessments + evidence pack — PROPOSED USAGE

Versioned review packs and a strict preflight CLI are implemented. Hosted CI execution, OTA integration and production billing are not claimed.

Custom profile onboarding — ONE-OFF

Bringing an OEM's own board and bootloader fork into the registry, with acceptance-gated promotion to SUPPORTED.

Land one technical pilot — PLAN

Verify one OEM board/bootloader/change, then make repeatable preflight evidence part of that supported product line's release process.

Packaging and willingness to pay are hypotheses. Fleet size informs risk exposure, not per-device billing. No customers, measured ROI or combined global TAM are claimed.

All economic inputs on this slide are hypothetical. Technical proof: historical wolfBoot run a39610, 8/8 tests executed and 61 hashed artifacts. Build/emulation: 42 s; not an end-to-end AI latency promise.

COMPETITIVE LANDSCAPE

Inventory what changes. Verify the migration path. Then deliver with OTA.

Category · examples	Main documented focus	Relationship to ReleaseProof
Binarly · ONEKEY crypto inventory / PQC readiness	Binary analysis, dependencies, crypto exposure; Binarly also documents reachability and CI gates	Complementary inputs. No claim that they cannot integrate dynamic tests.
Finite State adjacent firmware release-assurance competitor	Product security, evidence-linked AI, release policy gates and audit-ready evidence	Closest adjacent positioning. We focus on cross-generation bootloader migration, not unique ownership of evidence gates.
Memfault · Mender · Foundries.io OTA / fleet lifecycle	Deployment, telemetry, staged rollout and secure-update workflows	Potential integration partners. ReleaseProof is a preflight check, not a delivery replacement.
Renode / Antmicro · upstream CI embedded emulation / testing building blocks	Repeatable execution and bootloader regression tests	We build on them with transition rules, negative tests, evidence projection and supervised AI.
ReleaseProof two supported nRF52840 configurations	Deployed → candidate tests; separate candidate/transition verdicts; conservative UNKNOWN handling	Scoped engineering assurance before rollout—not hardware validation, OTA or certification.

Next validation—not existing capabilities

Hardware-in-the-loop and interrupted-update tests, then commercial forks, STM32/NXP profiles and OTA/CBOM integrations. Each new configuration needs its own adapter, runner and acceptance.

Looking for industrial sensor design partners

Don't ask only whether your new firmware works. Ask whether deployed devices can get there—before rollout.

Public vendor documentation, Oct 2026: binarly.io, onekey.com, finitestate.io, memfault.com, mender.io, docs.foundries.io, renode.io. Sources in submission.md. Focus comparison only: no head-to-head benchmark, exclusivity or capability-absence claims.